

How Local Incentives Shape Conflict Network Formation: An Agent-Based Study of the Myanmar Civil War

Shaping Asia's Future: Development, Innovation, and Society in Transition

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Years
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- The Myanmar Civil War was triggered by the military coup in February 2021, when the armed forces overthrew the civilian government.
- The country has been divided into multiple factions engaged in prolonged armed conflicts.
- It is now one of the most severe ongoing conflicts in Asia, with tens of thousands of casualties and millions displaced.
- Because the conflict is highly decentralized, and the network of alliances and rivalries is constantly changing, its resolution remains uncertain.

Problem Statement

One way to study such conflicts is through network analysis, which helps us understand how actors interact with each other.

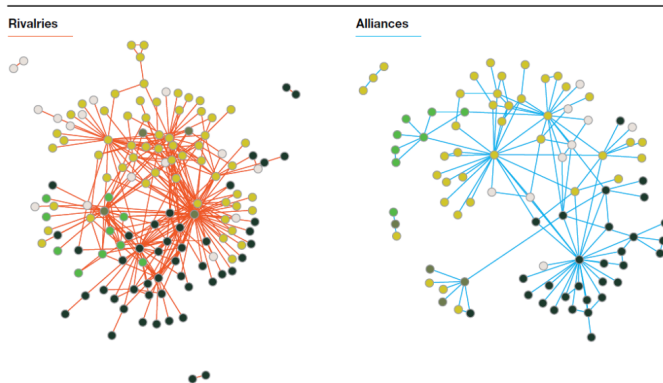


Figure: Visualized conflict networks. [3]

Analytical approach

Requires restrictive assumptions

- Homogeneous agents
- Perfect and globally observed information
- Static equilibrium

Agent-Based Modeling

Can operate without restrictions

- Heterogeneous agents
- Local incentives, incomplete information
- Dynamic evolution

- ① How do local incentives and limited information shape the formation of alliances and rivalries?
- ② Why stable yet inefficient conflict structures emerge?
- ③ Which factors most strongly influence conflict persistence, such as natural resources, territorial control, and human resources?

Inspired by the König et al.(2017) - "Networks in Conflict: Theory And Evidence From the Great War Of Africa"

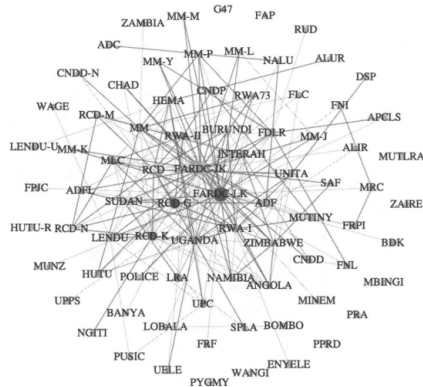


Figure: Visualization of conflict network of Second Congo War [2]

König et al.(2017) model:

- ➊ Start with a fixed network of alliances and enmities.
- ➋ Embed the network into strategic model, where each group maximizes its payoff.
- ➌ Solve for equilibrium conditions and obtain each actor's fighting effort.
- ➍ Use fighting efforts to estimate fighting externalities and to inform policy decision.

However, the model relies on strong analytical assumptions:

- ① Equilibrium assumption
- ② Network is static

To bypass these constraints, we propose a computational approach that captures decentralized interactions and local incentives.

We approximate the shape of Myanmar using a square grid (spatial discretization method).



Figure: Map of Myanmar [1]



Figure: Map of Myanmar, grid approximation

Methodology

- 1 Select five most influential actors in the conflict based on their access to natural resources, manpower, and control of fertile land.
- 2 Place these actors on the grid according to their actual geographical location
- 3 Allocate resources to the tiles using publicly available data.

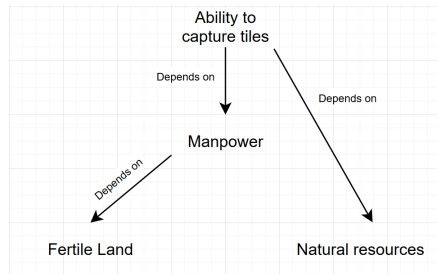


Figure: Resource interdependence diagram

The initial enmity and friendship networks are computed using ACLED data.

- Consider only "Battle" and "Strategic development" events.
- Each record includes time, location, and participants of events on both sides.
- Obtain list of all participants and iterate through it.

	event_date	event_type	actor1	assoc_actor_1	actor2	assoc_actor_2
18418	2023-01-01	Battles	KNU/KNLA: Karen National Union/Karen National ...	Lion Battalion Commando Special Force; PDF: Pe...	Military Forces of Myanmar (2021-) Border Guar...	Military Forces of Myanmar (2021-)
18432	2023-01-01	Battles	SBTF: Salingyi Border Task Force	Unidentified Anti-Coup Armed Group	Military Forces of Myanmar (2021-)	Police Forces of Myanmar (2021-)
18433	2023-01-01	Battles	Kanbalu UG: Kanbalu Underground Warriors	MPF U-18: Myanmar Paladin Force U-18; Myay Pyi...	Military Forces of Myanmar (2021-)	Pyu Saw Htee
18445	2023-01-01	Battles	People's Defense Force - Khin-U	People's Defense Force - Shwebo District Batta...	Military Forces of Myanmar (2021-)	Pyu Saw Htee
18449	2023-01-02	Battles	KNU/KNLA: Karen National Union/Karen National ...	Lion Battalion Commando Special Force; PDF: Pe...	Military Forces of Myanmar (2021-)	Military Forces of Myanmar (2021-) Border Guar...

Figure: Snapshot of event dataset from ACLED

Iterate through the list of actors and observe their interaction over time:

- If two actors have fought against each other at least once and never fought on the same side, they are connected with an edge in the enmity network.
- If two actors have fought together at least once and never opposed each other, they are connected with an edge in the friendship network.

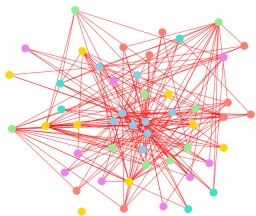


Figure: Enmity network of actors in Myanmar Civil War (2021 - Present)

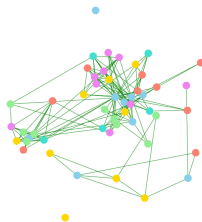


Figure: Friendship network of actors in Myanmar Civil War (2021 - Present)

Methodology

Once the model runs, each agent aims to occupy the tile within its vision radius that maximizes its utility. Consider agent A who wants to occupy such a tile:

- If the tile is not contested, the agent occupies it, pays a fixed movement cost, and claims its resources.
- These resources are then shared among Agent A's allies. The more powerful allies receive a larger share of resources.
- If maintaining an alliance is too costly, Agent A may decide to leave it.

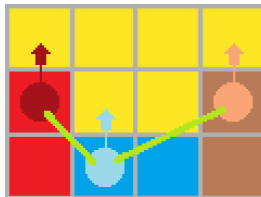


Figure: Agent A (blue) occupying free tile while in alliance with other two

Once the model runs, each agent aims to occupy the tile within its vision radius that maximizes its utility. Consider agent A who wants to occupy such a tile:

- If another agent B decides to contest a tile or already occupies it, then the conflict occurs, and agents are connected by an enmity edge.
- The cost of conflict for both agents depends on their manpower, natural resources and alliance network.
- To improve its chances, agent A might consider forming an alliance with others, even if it means sharing the resources of a contested tile.

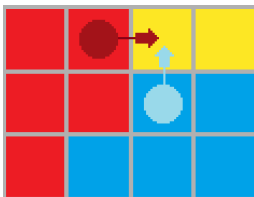


Figure: Agents contesting for a tile

- It's worth noting that agents A and B don't have complete information about each other.
- One of them might decide to engage in conflict and still lose.
- The cost of losing is shared among its allies, which further incentivizes alliance formation.

Results

At the current stage of the project, we have obtained several preliminary results. They demonstrate the feasibility of the approach and provide initial insights into conflict network structure.

- First, using ACLED data, we constructed the conflict network of the Myanmar Civil War, which includes over 30 different actors.
- This will serve as the basis for our initial network setup in the agent-based simulation.

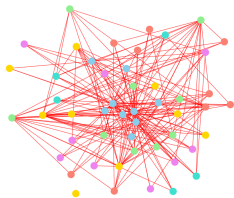


Figure: Enmity network of actors in Myanmar Civil War (2021 - Present)

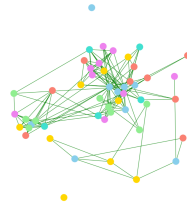


Figure: Friendship network of actors in Myanmar Civil War (2021 - Present)

In the previous project, we analyzed the conflict structure of the War in Darfur in Sudan:

- Constructed the conflict network.
- Performed centrality analysis of the network to identify the most important participants.
- Ranking is based on their network position rather than resource endowment.
- Some actors gain their influence from leveraging their connections instead of relying on resources.

	Degree centrality	Closeness centrality	Betweenness centrality	Katzs prestige	Eigenvector centrality	Bonacich centrality	Katz-Bonacich centrality
Military Forces of Sudan (1989-2019)	0.28	0.6111111	0.1908571429	-1.990989e-02	5.878990e-01	-0.16650006	154.2337624
Unidentified Armed Group (Sudan)	0.22	0.5689655	0.1802643343	-1.564349e-02	4.470220e-01	-0.52929684	146.3245452
SLM/A: Sudan Liberation Movement/Army	0.18	0.5322581	0.1307133139	-1.279922e-02	4.201220e-01	-1.00170917	112.2958169
JEM: Justice and Equality Movement	0.18	0.5156250	0.0915568513	-1.279922e-02	4.502306e-01	-0.58076378	90.7069810
Janjaweed	0.12	0.4852941	0.0426666667	-8.532810e-03	3.604076e-01	-0.13742576	1.4912127

Figure: Centrality ranking of Sudan conflict

We developed an agent-based simulation where we explored the evolution of the network using distance-based utility.

- The model imitates decentralized network formation
- We examined the extent to which it converged to network topologies predicted by analytical methods.

The integration of all these components into a unified model of the Myanmar conflict is still in progress. Our work focuses on two directions:

- ① Econometric analysis of how armed conflicts affect resource expenditure among competing actors.
- ② Implementation of spatial discretisation methods for grid-based setup.

This framework allows us to study the evolution of the network by varying parameters of the unified model and compare it with real-world conflict dynamics. We can investigate:

- ① Which agents, despite limited resources, significantly contributed to kickstarting the conflict.
- ② Which types of resources most strongly influenced the evolution of the conflict network, and why.
- ③ Which actors are likely to align or compete in the future, and what incentives contribute to these decisions.

In addition, this framework allows us to:

- ➊ Identify regions that require targeted policy attention by analyzing which areas in the model are most frequently contested.
- ➋ Identify which incentives could be used to reduce the persistence of conflict networks.

We propose a framework that combines empirical network analysis and agent-based modeling. Our preliminary results show that this approach is feasible and capable of capturing the structure of real-world conflicts.

We focus integrating these separate components into a fully calibrated model of the Myanmar Civil War.

The goal is to:

- ① Better understand conflict dynamics
- ② Understand which incentives drive them most strongly.
- ③ Inform potential intervention strategies that may contribute to conflict resolution or de-escalation.

-  Xinwei Zhang Adrienne Nenow and Melanie Salzarulo.
Current issues in refugee education.
-  Michael D. König, Dominic Rohner, Mathias Thoenig, and Fabrizio Zilibotti.
Networks in conflict: Theory and evidence from the great war of africa.
Econometrica, 85:1093–1132, 2017.
-  University of Florida; College of Liberal Arts and Sciences.
Conflict networks in north and west africa.

Thank You!